

TrES-5b

R-2323

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-5_150802 · created 2026-08-18T01:12:50+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457236.7684 +/- 0.0019 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.183 +/- 0.053
Transit depth (Rp/Rs)^2	3.36 +/- 1.95 [%]
Orbital Inclination (inc)	86.89 +/- 2.86
Airmass coefficient 1 (a1)	35.75 +/- 19.96
Airmass coefficient 2 (a2)	-2.97 +/- 0.69
Scatter in the residuals of the lightcurve fit is	59.73 %
Best Comparison Star	None
Optimal Aperture	16.02
Optimal Annulus	37.85
Transit Duration (day)	0.084 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.183 $\pm$ 0.053 vs 0.142 (deviation 0.53 σ)
Duration fit vs published	0.084 d vs 0.0758 d (deviation 0.51 σ)
Mid-time O-C	-4.7 min
Depth SNR	2.4
Residual scatter	59.73 %

Light curve

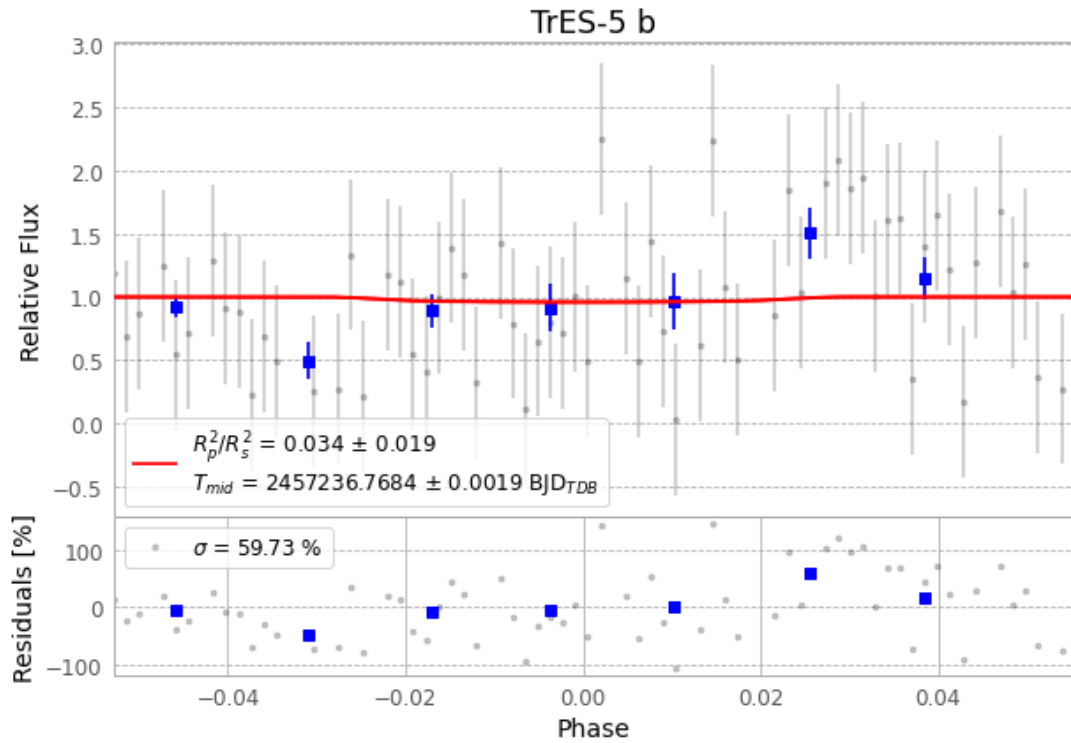


Figure 1 — FinalLightCurve_TrES-5 b_2015-08-01.png (SHA-256 130d6168159e8529...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [219, 346], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 45c0d1396fcedf80...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

78 FITS frames (51 MB), TRES-5150802043012.FITS → TRES-5150802082112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-150802.log (ace35921d10a...), FinalLightCurve_TrES-5 b_2015-08-01.png (130d6168159e...), FinalLightCurve_TrES-5 b_2015-08-01.csv (9cde0e0d0297...)

Compute resources

Wall time 231.0 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-18 01:12Z.